

Mobile App Testing – Appium

Testing Approach

This project focuses on **mobile application functional and UI testing** using **Appium, Java, Selenium WebDriver, and TestNG**.

The testing approach included:

- Creating and executing manual test scenarios before automation.
- Automating key mobile application workflows using Appium.
- Using **UiAutomator2** for Android automation.
- Using explicit waits to handle mobile application elements and synchronization.
- Validating navigation between application screens.
- Testing login functionality with both valid and invalid credentials.
- Validating login page elements, layout, and alignment.
- Testing input validation, including invalid email and password scenarios.
- Testing boundary and negative scenarios, such as passwords exceeding the allowed maximum length.
- Recording test execution results as **PASS / FAIL**.
- Reporting defects when the actual application behavior did not match the expected result.

Test Coverage

The automated tests covered:

Test Case	Description	Result
TC-001	Validate Menu Opens Successfully	PASS
TC-002	Login with Valid Credentials	PASS
TC-003	Validate Navigation to Login Page from Menu	PASS
TC-004	Validate Login Page Layout and Alignment	PASS
TC-005	Validate Login with Password Exceeding Maximum Length	FAIL

Key Findings and Insights

- The application successfully handled valid and invalid login scenarios.
- Invalid email and invalid password inputs were correctly rejected.
- The Menu navigation and Login page navigation worked as expected.
- The Login page layout and element alignment were successfully validated.
- A validation defect was identified where the application **accepted a password longer than the maximum allowed length of 8 characters**.
- The defect was documented with reproduction steps, expected result, actual result, severity, and priority.
- The project demonstrated the importance of testing both **positive and negative scenarios**, rather than testing only successful user flows.
- Automation helped provide repeatable test execution and clear PASS/FAIL results through TestNG.

Test Execution

Automated tests were executed using **TestNG**.

Example execution result:

Total tests run: 4

Passes: 4

Failures: 0

Skips: 0

The failed password-length validation was documented separately as a defect and linked to the corresponding test case.

Tools & Technologies

- Java
- Appium
- Appium Java Client
- Selenium WebDriver
- TestNG
- UiAutomator2
- Android Emulator
- IntelliJ IDEA
- Git / GitHub

Conclusion

This project demonstrates a practical approach to **Android mobile application testing**, combining functional testing, negative testing, UI validation, test automation, and defect reporting.